Berke Göktuğ Sarılı

Automation & SCADA Engineer | Electrical & Electronics Engineer

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PROFESSIONAL SUMMARY

Automation and SCADA engineer: two years in automation (custom hydraulic presses, then waste-to-energy, biogas, medical waste and wastewater plants) plus a year as an electrical engineer officer responsible for a high-voltage installation. Hands-on from EPLAN design and Siemens PLC programming (S7-300/1200/1500) to commissioning, live retrofits and troubleshooting. Builds the plant data layer end to end — PLC → KEPServerEX → SQL Server / PostgreSQL → dashboards and machine-learning-based predictive maintenance — and ships production web applications that digitalize field operations, including a live GPS fleet-tracking and route-optimization platform (rotadoa.com).

CORE SKILLS

PLC & Control	Siemens TIA Portal (S7-300/400/1200/1500, ET 200SP), LAD/SCL/GRAPH, PID control, safety logic (light curtains, E-stop, safety encoders), servo & drive commissioning (SINAMICS G120/V20, ABB), Sinumerik CNC
SCADA & HMI	WinCC Runtime Professional, alarm management, recipes, trending, automated reporting, WebNavigator / WebUX remote access
Industrial Data & Comms	Kepware KEPServerEX, OPC UA/DA, PostgreSQL, Microsoft SQL Server, Grafana, PROFINET, PROFIBUS, Modbus RTU/TCP, S7 PUT/GET, Siemens SCALANCE (S615, M874-3), wireless remote I/O, industrial cybersecurity
Software	Python (machine learning, data analysis), TypeScript/JavaScript, React, Node.js (Express), REST, WebSocket, SQL, Prisma, Docker, Linux, Git
Electrical Design	EPLAN Electric P8, panel design, high-voltage operation responsibility (EMO certified), Proteus, Eagle (PCB)
Languages	Turkish (native), English (C1)

PROFESSIONAL EXPERIENCE

Automation Engineer — ITC Invest Trading & Consulting AG

Ankara, Türkiye | February 2026 – Present | Integrated waste management and waste-to-energy group (biogas, incineration, medical waste, wastewater treatment) — multiple plants across Türkiye

- Own automation support for multiple plants: Siemens S7-1500 programming, WinCC RT Professional SCADA development and on-call troubleshooting (encoders, safety/STO circuits, PROFINET device integration, CPU communication failures).
- Re-engineered a medical waste sterilization machine end to end: wrote the complete PLC/SCADA control software from scratch (TIA Portal V19, ET 200SP, WinCC RT Pro), implemented the sterilization cycle logic, added an automated date-stamped PDF cycle-report archive for audit traceability, moved report data to PostgreSQL, and performed a 12-finding safety review of the logic.
- Performed a live CPU migration at an operating plant (CPU 1510SP-1 PN → CPU 1515-2 PN with CM 1542-5 + IM 155-6) after the controller ran out of memory, then restored the full KEPServerEX data feed.
- Built PLC → KEPServerEX → SQL Server/PostgreSQL data pipelines across sites: energy analyzers, conveyor run-hour and direction counters, carbon filter data, heat-meter (calorimeter) data; commissioned a SCALANCE S615 security router for a remote site.
- Developed machine-learning predictive maintenance scoring for 300+ equipment items across 7 process modules with Grafana dashboards (gearbox failure-time estimation, gas-sensor fault prediction); added RTD cards to bring gearbox temperatures into SCADA.
- Designed, built and launched rotadoa.com, the company's GPS fleet-tracking and route-optimization platform for recycling collection vehicles: traffic-aware routing with alternatives (TomTom / OSRM fallback), real-time multi-vehicle tracking over WebSocket, offline GPS queueing, map-matching of real tracks, driver Android app, capacity-based automatic discharge stops, fuel-cost and planned-vs-actual reporting. Stack: React + TypeScript PWA, Node.js, Prisma, Docker.
- Developed a facility inspection application replacing paper control forms (KKS-coded checklists, photo evidence, weighted scoring); rolled out to 5 plants and extended to the IT and process departments.

- Integrated third-party package units into plant SCADA (S7 PUT/GET for solid-polymer dosing units at two plants), deployed WebNavigator remote SCADA access, and tested the company's first wireless remote I/O module on a moving crane with zero data loss.
- Root-caused chronic SCADA client slowness across multiple plants (network adapter / DNS configuration) and standardized the fix; authored 40+ internal training guides (SCADA, PostgreSQL, PID, drives, remote access) and delivered PostgreSQL training.

Engineering Lieutenant (Military Service) — Turkish Air Force, Akıncı Air Base
Ankara, Türkiye | November 2024 – November 2025 | Electrical Engineer Officer

- Managed the air base's high-voltage infrastructure: transformer substation maintenance, lightning protection systems, and electrical surveys and acceptance of new buildings, in compliance with military standards and safety regulations.
- Coordinated technical teams in field operations.

Automation Engineer — Dirinler Machinery
İzmir, Türkiye | July 2023 – November 2024 | Industrial hydraulic presses & CNC systems

- Designed (EPLAN) and programmed (Siemens S7-300/1200/1500) H-type and C-type hydraulic presses from 63 to 1500 tons; each order was a custom machine, engineered and commissioned individually.
- Delivered machines for domestic and international customers, including 630-ton and other presses exported to Ukraine, with English/Russian user manuals.
- Integrated and commissioned servo-driven manipulators, transfer systems, die clamping, automatic stroke control, heating panels and safety encoders; implemented light curtains, emergency stop circuits and two-hand control in PLC safety logic.
- Developed PLC software for 3–5 axis gantry CNC machines (Sinumerik); prepared quality documentation, user manuals, tonnage test reports and safety reports.

R&D Engineer (Defense Industry) — KZ Mechatronics
Balıkesir, Türkiye | January 2023 – April 2023

- PCB design (Eagle, Proteus) and embedded software (STM32, Arduino) for security and monitoring devices; prototype validation in field and lab tests.

EDUCATION

B.Sc. in Electrical & Electronics Engineering — Balıkesir University
2019 – 2023 | GPA 3.39 / 4.00 | Erasmus+ exchange: Politechnika Częstochowska, Poland (2021 – 2022)

- Graduation project: LoRaWAN-based PLC communication (up to 15 km, no internet) — S7-1200 via Modbus transferring remote boiler-room temperatures and driving a fire-sprinkler control loop.
- Delivered PLC laboratory training to students.

CERTIFICATES & TRAINING

Operation Responsibility in High-Voltage Facilities — Chamber of Electrical Engineers (EMO), 2025 · Industrial Communication & Cybersecurity in Automation — 2025 · PLC Training, IoT — Ministry of National Education, 2025 · EPLAN — 2025 · Object-Oriented Programming with Python — Gazi University, 2025 · Authorized Boiler Operator License — 2025 · English C1 — Erasmus+, 2020

ADDITIONAL

Work & Travel, Virginia, USA (2022): sales and customer relations in an English-speaking environment.